

Biomolecules



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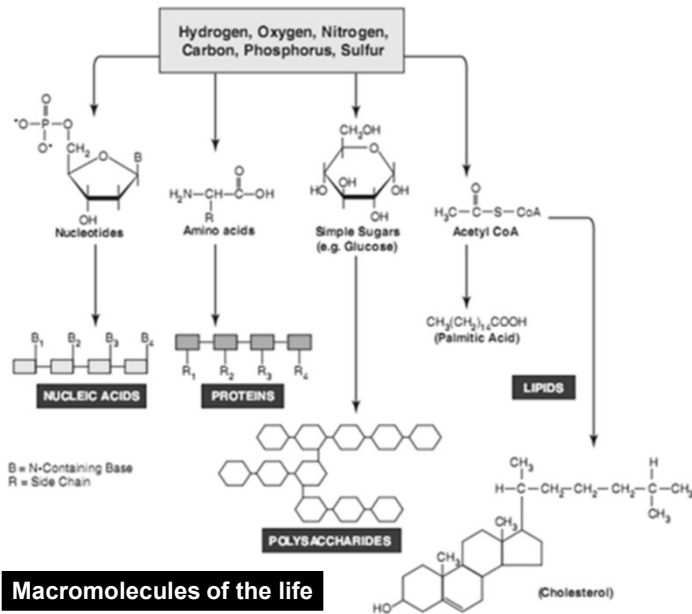
Biomolecules

- ❖ Biomolecule, also called biological molecule, any of the substances that are produced by cells and living organisms. Biomolecules have a wide range of **sizes** and **structures** and perform a vast array of **functions**. The four major types of biomolecules are **carbohydrates**, **lipids**, **nucleic acids**, and **proteins**.
- ❖ A biomolecule is a loosely used term for molecules present in organisms that are essential to one or more typically biological processes, such as **energy generation**, **cell division**, **morphogenesis**, or **development**.
- ❖ They are an organic molecule and are important for the survival of living cells. Some of valuable biomolecules have huge demand, which cannot be fulfilled from their renewable resources. Microbes have been used as a cell factory for their alternative production. The strategies adopted for engineering a high yielding biomolecule producing microbe are overexpression, knockout, and activation of gene.

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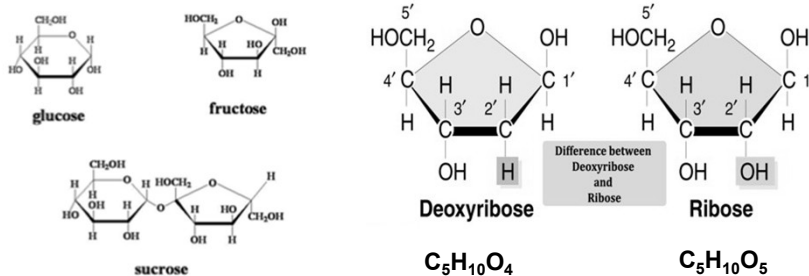
Biomolecules



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Carbohydrates

- ❖ Building blocks of the biosphere.
- ❖ Carbohydrates are polyhydroxy aldehydes and ketones or substances that yield one of these compounds on hydrolysis.
- ❖ They are composed of three elements: C, H, O in a ratio of 1:2:1 with a General Formula: $C_nH_{2n}O_n$ OR $(CH_2O)_n$; where n=3 or larger
- ❖ e.g. Glucose $C_6H_{12}O_6$



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Classification of carbohydrates

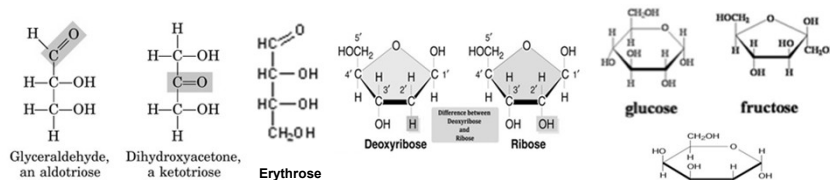
❖ Carbohydrates are classified according to the number of simple sugar monomer units present in the molecule (upon hydrolysis).

1. **Monosaccharide**
2. **Disaccharide**
3. **Oligisaccharide**
4. **Polysaccharides**

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Monosaccharides

- ❖ Mono, means "one", and saccharide, means "sugar". So, these are simple sugars having one sugar molecule OR "on hydrolysis these sugar break down into only one type of monomeric unit" (i.e. Glucose, Fructose and Galactose etc). No further hydrolysis is possible. These are colorless, crystalline compounds with sweet taste.
- ❖ Based upon the number of carbon atoms in a molecules, they can be classified as:
- **Trioses:** contain 3-carbon atoms e.g. Glyceraldehydes, Dihydroxy acetone
 - **Tetroses:** contain 4-carbon atoms e.g. Erythrose
 - **Pentoses:** contain 5-carbon atoms e.g. Ribose, Deoxyribose
 - **Hexose:** contain 6-carbon atoms e.g. Glucose, fructose, galactose
 - **Heptose:** contain 7-carbon atoms e.g. Sedoheptulose



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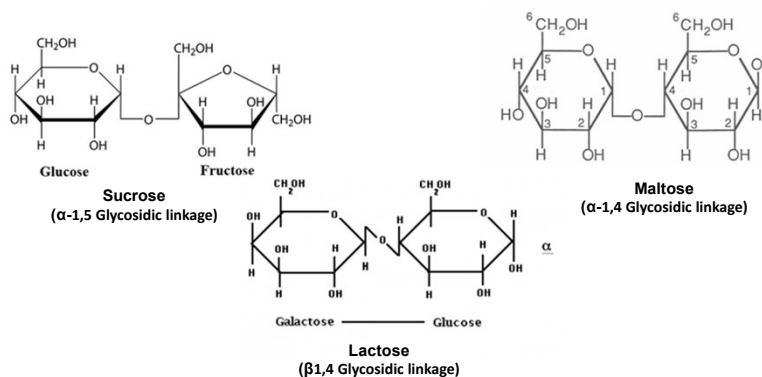
Importance of Monosaccharides

- ❖ Glyceraldehyde is the most common form of triose found in the living cells.
- ❖ Pentoses and Hexoses are the most abundant monosaccharides present in protoplasm.
- ❖ Glucose is the chief source of energy. All carbohydrates are ultimately converted to glucose in the body. Glucose serves as the respiratory substrate in all living cells.
- ❖ Ribose is the important component of RNA, NAD, NADP and AcetylCoA.
- ❖ DNA contains Deoxy ribose sugar, which is a pentoses.
- ❖ Ribulose is essential for photosynthesis.
- ❖ Erythrose is a tetrose involved in synthesis of lignin and anthocyanin.
- ❖ Fructose: It is found in cane sugar, fruit juices, honey and the nectar of flowers.
- ❖ Galactose is found commonly in the cells of red algae.

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Disaccharides

- ❖ Disaccharides are the molecules which on hydrolysis yield 2 monosaccharide units. Three most important disaccharides are **sucrose**, **lactose** and **maltose**.
- ❖ 2 monosaccharide units are linked through a **Glycosidic linkage** which is formed through a dehydration reaction that releases a water molecule. To break this bond, named a glycosidic bond, you add water, this reaction is named hydrolysis.



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Importance of Disaccharides

- ❖ **Maltose:** It is made of two glucose molecules. It is formed from starch during germination of barley or wheat. It is commonly called malt sugar.
- ❖ **Lactose:** is a disaccharide found in milk and hence commonly called milk sugar. It is formed by condensation of a glucose and a galactose.
- ❖ **Sucrose:** It is a disaccharide found extensively in plants. It is commonly called cane sugar. It is formed by a condensation of a molecule of glucose and a molecule of fructose.
- ❖ Raffinose is a trisaccharide formed by condensation of a molecule each of glucose, fructose and galactose. It occurs in plants.

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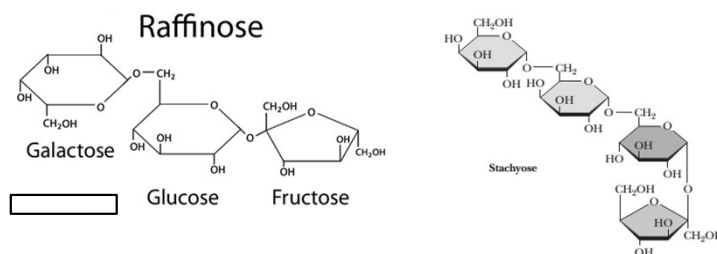
Oligosaccharides

- ❖ They are carbohydrates formed by joining of 3 - 10 monosaccharide molecules. The bond formed between monosaccharides, is called glycosidic bond.
- ❖ A large number of oligosaccharides have been prepared by partially breaking down more complex carbohydrates (polysaccharides).
- ❖ They can be hydrolyzed by acid to form constituent monosaccharide units.
- ❖ Oligosaccharides are classified according to the number of monosaccharide units (or monomers). They are composed of disaccharides, trisaccharides, tetrasaccharides, pentasaccharides and hexasaccharides.
- ❖ Among these, disaccharides are most abundant in the living system.

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Importance of Oligosaccharides

- ❖ The nondigestibility of oligosaccharides provides a health benefit as they help to feed friendly bacteria, also known as **probiotics**, in the large intestine. Therefore, oligosaccharides are often referred to as **prebiotics**.
- ❖ Two common oligosaccharides are **raffinose**, composed of 3-monosaccharide units, and **stachyose**, composed of 4-monosaccharide units, found in beans and other legumes.



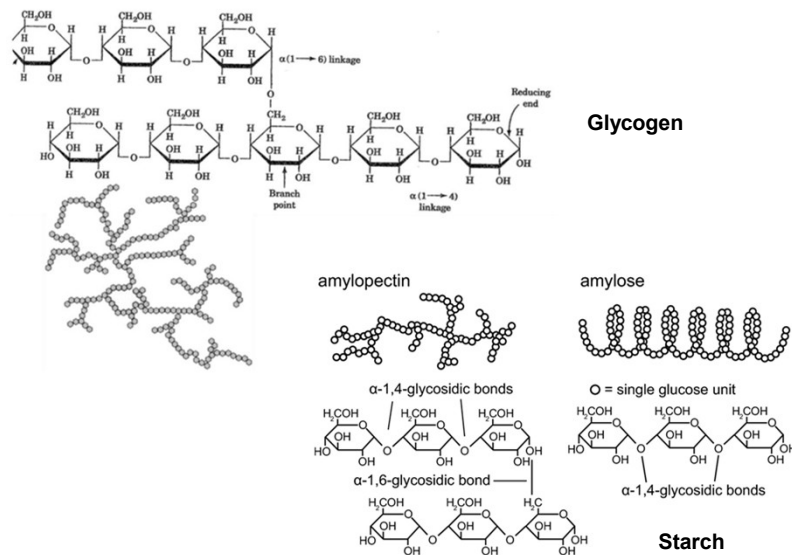
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Polysaccharides

- ❖ Those molecules which yield more than 10 monosaccharide units.
- ❖ Polysaccharides are polymeric carbohydrate structures, formed of repeating units (either mono- or di-saccharides) joined together by glycosidic bonds.
- ❖ These structures are often linear, but may contain various degrees of branching. Depending on the structure, these macromolecules can have distinct properties from their monosaccharide building blocks. They may be amorphous or even insoluble in water.
- **Homopolysaccharide**: Made up of similar monosaccharide units eg. starch, glycogen, cellulose, chitin and dextran.
- **Heteropolysaccharides**: Made up of dissimilar monosaccharide units eg. peptidoglycan and agarose.
- ❖ Polysaccharide is an important class of biological polymers. Their function in living organisms is usually either structure- or storage-related.

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Polysaccharides



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Polysaccharides

❖ Digestible Polysaccharides

The only digestible polysaccharide is starch, which is a large molecule composed simply of glucose. It is broken down first in the mouth and then in the small intestine by the enzyme amylase. All the starch that is absorbed into the body is in the form of individual glucose molecules, which are delivered to the cells by the blood for energy. Extra glucose is reassembled into the starch like molecule called glycogen to be stored in the liver and the muscles. Excessively high intake will be converted to fat. e.g. grains, root vegetables and legumes.

❖ Nondigestible Polysaccharides

Unrefined grains, starchy and nonstarchy vegetables, legumes and fruit provide various nondigestible polysaccharides called fiber, which fall into two main categories: Insoluble fiber, also known as cellulose, is abundant in wheat bran and vegetables. Cellulose, like starch, is composed purely of glucose molecules, but the bonds are such that human enzymes can't break them. Its main role is to keep undigested food moving through the intestines, preventing constipation. Soluble fibers, such as pectins, gums and mucilages, are abundant in oat bran, beans and many fruits.

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Polysaccharides

❖ **Storage polysaccharides**

Starch - Plants

Glycogen - Animals

❖ **Structural polysaccharides (Glycans)**

Cellulose - Plants

Chitin - Animals

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THANK YOU

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